DO NOW — RECALL

Programming Techniques · Functions and Procedures

Name: _____ Class: _____ Date: _____

1. What is the final result of x=5; x-=2;?

2. What does a binary search need first?

3. How many times runs for(i=0;i<4;i++)?

4. What does a truth table list?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

3

2. Model answer

Sorted input

3. Model answer

4

4. Model answer

Outputs for input combinations

By the end, you can…

• Explain returned values and void methods

• Trace parameter values

• Define and call a C# method

Vocabulary — know these terms

Recall the term — function

Recall the term — procedure

Recall the term — parameter

Recall the term — argument

Recall the term — return

NEW LEARNING

Read, annotate and explain

Return a computed value

static int Double(int value) { return value * 2; } int result = Double(6); Console.WriteLine(result);

Explain the important decision

The argument is copied into a local parameter. return sends a computed value to the caller; it does not itself print the value.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Return a computed value

Pass the argument

Double(6) starts a call with local parameter value=6. The caller waits for an integer result.

Calculate and return

value*2 evaluates to 12. return 12 ends this call and supplies that integer to the caller.

Use the returned value

The assignment stores 12 in result; Console.WriteLine(result) prints it. A void procedure can print, but supplies no value for this assignment.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Problem 1

A function adds a and b; what is Add(3, 4)?

Method / working:

2. Problem 2

A void method prints a greeting. Does it return a string?

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Apply independently
Write a C# method returning the cube of an integer. [4]

Question 2

Apply independently
Explain the purpose of a parameter. [2]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Apply independently
Double(n) returns n*2. Calculate Double(Double(2)) and show both returns. [2]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

True or false: a void C# method must return a calculated value.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.